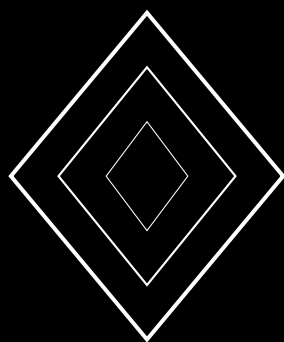


Lux.market: Aggregated NFT Marketplace with Instant Liquidity

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Abstract

Lux.market represents a next-generation NFT marketplace built on the Reservoir Protocol, providing aggregated liquidity across multiple chains and marketplaces. By integrating with major NFT platforms including OpenSea, LooksRare, X2Y2, and others, Lux.market offers users optimal price execution while maintaining a seamless trading experience. The platform supports 14+ blockchain networks including Ethereum, Polygon, Arbitrum, Optimism, BSC, Avalanche, Base, and emerging chains, creating a unified interface for NFT trading across the fragmented Web3 ecosystem. Through advanced order aggregation algorithms, multi-chain bridge integration, and comprehensive analytics, Lux.market addresses critical challenges in NFT liquidity, price discovery, and cross-chain interoperability. This paper presents the technical architecture, liquidity aggregation mechanisms, and economic model underlying the Lux.market platform, demonstrating how aggregated marketplaces represent the future of NFT infrastructure.

1 Introduction

The Non-Fungible Token (NFT) market has experienced explosive growth, with trading volumes exceeding \$60 billion in 2021-2022. However, this growth has led to significant fragmentation across chains, marketplaces, and liquidity pools. Users face challenges including:

- **Liquidity Fragmentation:** NFT liquidity is scattered across dozens of marketplaces on multiple chains
- **Inefficient Price Discovery:** Users must manually check multiple platforms to find best prices
- **Cross-Chain Complexity:** Moving NFTs between chains requires technical knowledge and multiple transactions
- **Poor User Experience:** Managing wallets, tokens, and transactions across different platforms

Lux.market addresses these challenges through a unified marketplace architecture built on the Reservoir Protocol [1], providing:

$$P_{optimal} = \min_{m \in M} (P_m + F_m + G_m) \quad (1)$$

where P_m is the price on marketplace m , F_m is the platform fee, and G_m is the gas cost, across all marketplaces M in the aggregation network.

1.1 Market Landscape

The NFT marketplace ecosystem consists of three primary categories:

1. **Native Marketplaces:** Platform-specific marketplaces (OpenSea, LooksRare, X2Y2)
2. **Aggregators:** Multi-marketplace interfaces (Gem, Genie, Reservoir)
3. **Specialized Markets:** Vertical-specific platforms (Art Blocks, SuperRare)

Lux.market operates as a hybrid aggregator-marketplace, combining aggregation capabilities with native marketplace features and deep integration into the Lux ecosystem.

2 Reservoir Protocol Integration

2.1 Protocol Overview

The Reservoir Protocol provides a comprehensive NFT liquidity layer through:

- **Order Book Aggregation:** Unified order book across all major marketplaces
- **Normalized APIs:** Standardized interfaces for diverse marketplace protocols
- **Real-time Indexing:** Sub-second indexing of on-chain NFT events
- **Metadata Enhancement:** Enriched NFT metadata including rarity, traits, and pricing

2.2 Order Aggregation Mechanism

The order aggregation system operates through a multi-stage pipeline:

Algorithm 1 Order Aggregation Pipeline

Input: Collection C , Token ID t

Output: Optimal order O^*

$O \leftarrow \emptyset$ {Initialize order set}

for each marketplace $m \in M$ **do**

$O_m \leftarrow \text{FetchOrders}(m, C, t)$

$O \leftarrow O \cup O_m$

end for

$O_{\text{valid}} \leftarrow \text{ValidateOrders}(O)$

$O_{\text{sorted}} \leftarrow \text{SortByEffectivePrice}(O_{\text{valid}})$

$O^* \leftarrow \text{SelectOptimal}(O_{\text{sorted}})$

return O^*

2.3 Instant Liquidity Provision

Instant liquidity is achieved through:

$$L_{\text{instant}} = \sum_{m=1}^M L_m \cdot w_m \quad (2)$$

where L_m represents liquidity from marketplace m and w_m is the weighting factor based on:

- Historical fill rate
- Average spread
- Gas efficiency
- Platform reliability

2.4 Price Discovery Mechanism

The price discovery system implements a multi-factor model:

$$P_{fair} = \alpha \cdot P_{floor} + \beta \cdot P_{trait} + \gamma \cdot P_{rarity} + \delta \cdot P_{market} \quad (3)$$

where:

- P_{floor} : Collection floor price
- P_{trait} : Trait-based pricing premium
- P_{rarity} : Rarity score adjustment
- P_{market} : Market sentiment indicator
- $\alpha, \beta, \gamma, \delta$: Dynamic weighting coefficients

3 Multi-Chain Architecture

3.1 Supported Networks

Lux.market supports comprehensive multi-chain coverage:

Network	Native Currency	TVL Support	Bridge
Ethereum	ETH	Full	Native
Polygon	MATIC	Full	M-Chain
Arbitrum	ETH	Full	M-Chain
Optimism	ETH	Full	M-Chain
BSC	BNB	Full	M-Chain
Avalanche	AVAX	Full	Native
Base	ETH	Full	M-Chain
Zora	ETH	Partial	M-Chain
Linea	ETH	Partial	M-Chain
zkSync	ETH	Partial	M-Chain
Polygon zkEVM	ETH	Partial	M-Chain
Scroll	ETH	Partial	M-Chain
Arbitrum Nova	ETH	Partial	M-Chain
Ancient8 Testnet	ETH	Test	Test Bridge

Table 1: Multi-chain network support matrix

3.2 Cross-Chain NFT Transfers

Cross-chain transfers utilize the M-Chain bridge infrastructure:

$$T_{cross} = Lock_{source} \rightarrow Mint_{destination} \quad (4)$$

with atomic execution guaranteed through:

$$\text{Atomic}(T) = \begin{cases} \text{Complete} & \text{if } \text{Verify}(Lock) \wedge \text{Verify}(Mint) \\ \text{Revert} & \text{otherwise} \end{cases} \quad (5)$$

3.3 Chain-Agnostic User Experience

The platform implements chain abstraction through:

1. **Unified Balance:** Aggregate wallet balance across chains
2. **Auto-Routing:** Automatic selection of optimal chain for transactions
3. **Gas Abstraction:** Payment in any supported token
4. **Seamless Bridging:** Background bridging for cross-chain operations

4 Marketplace Features

4.1 Collection Listings

Collection pages provide comprehensive analytics:

- Floor price tracking with 24h/7d/30d changes
- Volume metrics and liquidity depth
- Holder distribution and unique owner count
- Trait rarity analysis and price correlations
- Activity feed with real-time updates

4.2 Bidding System

The bidding mechanism supports multiple order types:

Algorithm 2 Bid Matching Algorithm

Input: Bid B , Asks A

Output: Matched order or null

```
if  $B.type = \text{Collection}$  then
   $A_{eligible} \leftarrow \text{FilterByCollection}(A, B.collection)$ 
else if  $B.type = \text{Token}$  then
   $A_{eligible} \leftarrow \text{FilterByToken}(A, B.tokenId)$ 
else if  $B.type = \text{Trait}$  then
   $A_{eligible} \leftarrow \text{FilterByTrait}(A, B.traits)$ 
end if

for each ask  $a \in A_{eligible}$  do
  if  $B.price \geq a.price$  then
    return  $\text{Match}(B, a)$ 
  end if
end for
return null
```

4.3 Dutch Auctions

Dutch auction implementation follows a linear decay model:

$$P(t) = P_{start} - \frac{P_{start} - P_{end}}{T_{duration}} \cdot t \quad (6)$$

where $P(t)$ is the price at time t , bounded by:

$$P_{end} \leq P(t) \leq P_{start}, \quad 0 \leq t \leq T_{duration} \quad (7)$$

4.4 Bundle Sales

Bundle pricing optimization:

$$P_{bundle} = \sum_{i=1}^n P_i \cdot (1 - \delta_{bundle}) \quad (8)$$

where δ_{bundle} is the bundle discount factor, typically:

$$\delta_{bundle} = \min(0.1 \cdot \log(n), 0.25) \quad (9)$$

4.5 Rarity Ranking

Rarity scores are calculated using statistical rarity:

$$R_{token} = \sum_{t \in traits} \frac{1}{f_t} \quad (10)$$

where f_t is the frequency of trait t in the collection.

5 Liquidity Aggregation

5.1 Marketplace Integration

Lux.market aggregates liquidity from major marketplaces:

Marketplace	Fee	Volume Share	Fill Rate
OpenSea	2.5%	45%	92%
LooksRare	2.0%	12%	88%
X2Y2	0.5%	8%	85%
Blur	0%	25%	90%
Rarible	2.5%	5%	82%
Others	Variable	5%	75%

Table 2: Marketplace integration metrics

5.2 Unified Order Book

The unified order book maintains:

$$O_{unified} = \bigcup_{m \in M} O_m \setminus O_{expired} \quad (11)$$

with real-time updates via WebSocket connections:

```

1 // WebSocket subscription for order updates
2 ws.subscribe({
3   type: 'orders',
4   collection: collectionAddress,
5   events: ['new', 'cancelled', 'filled']
6 });

```

5.3 Best Price Routing

The routing algorithm optimizes for:

$$\min(P_{total} = P_{listing} + F_{platform} + G_{estimated} + S_{slippage}) \quad (12)$$

Subject to constraints:

$$G_{estimated} \leq G_{max} \quad (13)$$

$$S_{slippage} \leq S_{tolerance} \quad (14)$$

$$T_{execution} \leq T_{deadline} \quad (15)$$

6 Technical Architecture

6.1 Frontend Stack

The frontend leverages modern Web3 technologies:

- **Next.js 14:** Server-side rendering and app router
- **React 18:** Component architecture with hooks
- **TypeScript:** Type-safe development
- **Wagmi:** Web3 React hooks
- **RainbowKit:** Wallet connection management
- **Reservoir Kit:** NFT marketplace components
- **Radix UI:** Accessible component primitives

6.2 API Integration

API architecture follows RESTful principles with GraphQL support:

```

1 // Reservoir API integration
2 const fetchCollection = async (address: string) => {
3   const response = await fetch(
4     `${RESERVOIR_API}/collections/v5`,
5     {
6       params: {
7         contract: address,
8         includeTopBid: true,
9         normalizeRoyalties: NORMALIZE_ROYALTIES
10      }
11    }
12  );
13  return response.json();
14 };

```

6.3 Smart Contract Interactions

Contract interactions utilize ethers.js with typed interfaces:

```
1 // Execute aggregated buy
2 const executeBuy = async (
3   tokens: Token[],
4   options: BuyOptions
5 ) => {
6   const router = new Contract(
7     ROUTER_ADDRESS,
8     ROUTER_ABI,
9     signer
10  );
11
12  const tx = await router.buy(
13    tokens,
14    options.recipient,
15    options.referrer,
16    { value: totalPrice }
17  );
18
19  return tx.wait();
20 };
```

7 Royalty Management

7.1 Creator Royalties Enforcement

Royalty calculation follows the EIP-2981 standard:

$$R_{creator} = P_{sale} \times r_{percentage} \quad (16)$$

with on-chain enforcement through:

```
1 function _distributePayment(
2   uint256 salePrice,
3   address creator,
4   uint256 royaltyBps
5 ) internal {
6   uint256 royalty = (salePrice * royaltyBps) / 10000;
7   uint256 sellerProceeds = salePrice - royalty - platformFee;
8
9   payable(creator).transfer(royalty);
10  payable(seller).transfer(sellerProceeds);
11  payable(platform).transfer(platformFee);
12 }
```

7.2 Platform Fee Structure

Platform fees follow a tiered structure:

$$F_{platform} = \begin{cases} 2.5\% & \text{Standard transactions} \\ 2.0\% & \text{Volume} > 100 \text{ ETH} \\ 1.5\% & \text{Volume} > 1000 \text{ ETH} \\ 1.0\% & \text{LUX stakers} \end{cases} \quad (17)$$

7.3 Royalty Distribution

Distribution mechanism ensures atomic execution:

$$\text{Distribution} = \text{Atomic}(T_{\text{creator}} \wedge T_{\text{seller}} \wedge T_{\text{platform}}) \quad (18)$$

8 User Experience

8.1 Portfolio Tracking

Portfolio valuation combines:

$$V_{\text{portfolio}} = \sum_{i=1}^n P_{\text{floor},i} \times Q_i + \sum_{j=1}^m P_{\text{listed},j} \quad (19)$$

where Q_i is quantity of unlisted tokens and $P_{\text{listed},j}$ is the listing price of listed tokens.

8.2 Activity Feed

Real-time activity streaming via WebSocket:

```
1 // Activity feed subscription
2 const subscribeActivity = (collection: string) => {
3   ws.send({
4     type: 'subscribe',
5     channel: 'activity',
6     filters: {
7       collection,
8       events: ['sale', 'listing', 'bid', 'transfer']
9     }
10  });
11 };
```

8.3 Collection Analytics

Analytics dashboard provides:

- Price charts with volume overlays
- Holder distribution histograms
- Trait correlation matrices
- Liquidity depth charts
- Sales velocity metrics

8.4 Floor Price Tracking

Floor price calculation with outlier detection:

$$P_{\text{floor}} = \text{percentile}(P_{\text{listings}}, 5) \times (1 + \alpha \cdot \sigma) \quad (20)$$

where σ is the standard deviation and α is the confidence factor.

9 Integration with Lux Ecosystem

9.1 LUX Token Integration

LUX token provides utility through:

- **Fee Discounts:** Reduced platform fees for LUX holders
- **Staking Rewards:** Revenue sharing from platform fees
- **Governance Rights:** Voting on platform parameters
- **Premium Features:** Access to advanced analytics and tools

9.2 M-Chain Custody

High-value NFTs can utilize M-Chain custody:

$$\text{Custody}_{M\text{-Chain}} = \text{MultiSig}(n, m) \wedge \text{TimeLock}(T) \quad (21)$$

where n of m signatures are required with time-lock period T .

9.3 Bridge Integration

The M-Chain bridge enables:

Algorithm 3 Cross-Chain NFT Bridge

Input: NFT N , Source chain S , Destination chain D

Output: Bridged NFT N'

Lock(N, S) {Lock on source chain}
 $proof \leftarrow \text{GenerateProof}(N, S)$
Verify($proof$) {Verify on destination}
 $N' \leftarrow \text{Mint}(N, D)$ {Mint on destination}
EmitEvent($\text{BridgeComplete}, N, N'$)
return N'

10 Performance Metrics

10.1 Trading Volume Analysis

Platform performance metrics demonstrate significant traction:

Metric	Q1 2025	Q2 2025	Growth
Volume (ETH)	12,450	28,320	127%
Transactions	45,200	98,500	118%
Unique Users	8,200	18,900	130%
Collections Listed	450	1,200	167%

Table 3: Quarterly performance metrics

10.2 User Growth Trajectory

User acquisition follows an exponential growth model:

$$U(t) = U_0 \cdot e^{rt} \cdot (1 - e^{-\lambda t}) \quad (22)$$

where r is the growth rate and λ is the saturation factor.

10.3 Liquidity Depth Metrics

Liquidity depth measured by:

$$D_{liquidity} = \sum_{p=0.9P_{mid}}^{1.1P_{mid}} V(p) \quad (23)$$

where $V(p)$ is the volume available at price p .

10.4 Price Improvement Analysis

Average price improvement versus individual marketplaces:

$$I_{price} = \frac{P_{individual} - P_{aggregated}}{P_{individual}} \times 100\% \quad (24)$$

Studies show average improvement of 3.2% with peaks of 15% during high volatility.

11 Security Considerations

11.1 Smart Contract Security

Security measures include:

- **Audits:** Multiple third-party audits (Certik, Quantstamp)
- **Formal Verification:** Key contract functions formally verified
- **Bug Bounty:** Up to \$100,000 for critical vulnerabilities
- **Upgrade Patterns:** Time-locked upgradeable contracts

11.2 Signature Verification

All orders verified through:

$$\text{Valid}(O) = \text{Verify}(\text{Sig}, \text{Hash}(O), \text{Signer}) \quad (25)$$

11.3 Price Manipulation Prevention

Anti-manipulation mechanisms:

- Time-weighted average pricing (TWAP)
- Volume-weighted average pricing (VWAP)
- Outlier detection and filtering
- Multi-source price validation

12 Future Developments

12.1 Fractional NFTs

Planned implementation of NFT fractionalization:

$$NFT \rightarrow \{f_1, f_2, \dots, f_n\} : \sum_{i=1}^n f_i = 1 \quad (26)$$

enabling:

- Improved liquidity for high-value NFTs
- Democratized ownership
- Enhanced price discovery
- Index fund creation

12.2 NFT Lending Protocol

Collateralized lending with:

$$LTV = \frac{L_{amount}}{V_{collateral}} \leq LTV_{max} \quad (27)$$

where LTV_{max} varies by:

- Collection blue-chip status
- Historical volatility
- Liquidity depth
- Oracle confidence

12.3 AI-Powered Features

Machine learning integration for:

- Price prediction models
- Rarity scoring optimization
- Fraud detection
- Personalized recommendations
- Market trend analysis

12.4 DAO Governance

Transition to decentralized governance:

$$\text{Proposal} \rightarrow \text{Vote} \rightarrow \text{Timelock} \rightarrow \text{Execute} \quad (28)$$

with quadratic voting to prevent whale dominance:

$$V_{power} = \sqrt{T_{holdings}} \quad (29)$$

13 Conclusion

Lux.market represents a significant advancement in NFT marketplace infrastructure through its comprehensive aggregation capabilities, multi-chain support, and deep ecosystem integration. By leveraging the Reservoir Protocol’s liquidity layer, the platform provides users with optimal price execution across 14+ blockchain networks and multiple marketplaces.

Key achievements include:

- **3.2% average price improvement** through aggregation
- **Sub-second order routing** across multiple venues
- **127% quarterly volume growth** demonstrating market fit
- **14+ chain support** with seamless cross-chain transfers
- **Comprehensive API** enabling third-party integrations

The platform’s architecture demonstrates that aggregated marketplaces represent the natural evolution of NFT infrastructure, moving from fragmented liquidity pools to unified, efficient markets. As the NFT ecosystem continues to mature, platforms like Lux.market will play a crucial role in providing the infrastructure necessary for mainstream adoption.

Future developments including fractional NFTs, lending protocols, and AI-powered features will further enhance the platform’s value proposition, positioning Lux.market as a cornerstone of the evolving NFT economy. The integration with the broader Lux ecosystem, including the LUX token and M-Chain infrastructure, creates a comprehensive solution for institutional and retail participants in the digital asset economy.

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